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(1) (1) Research Program/Project/Study Title: COLLEGE ADMISSION AND APTITUDE TEST DECISION SUPPORT SYSTEM	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:		PONTEVEDRA	
Province:		CAPIZ	
Region:		VI – WESTERN VISAYAS	
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☒ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Education Sector		(8) Discipline/Field of Specialization Information Technology	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

This study aimed to develop a College General Admission and Aptitude Test Decision Support System. This system project helps the university in every college to administer new freshmen students' examination to take a general admission and aptitude test as part of the requirement of the university. It can also help the students to decide for the course based on the result of the examination. It can also help department chair/ faculty to assist and advice students on the course they are most fitted to enroll.

This software has the following objectives: 1.) Design and develop the following system features: (a) System security feature to ensure that only authorized users can access the system; (b) Database feature that will store information provided by the system; (c) Automatic examination result computation; (d) Student course selection feature based on examination result; (e) Report query feature based on selected criteria; (f) Report generation feature for printing results and graphical results display of examination results; 2.) determine the acceptability level of College Admission and Aptitude Test Decision Support System: (a) Product Quality: (b) Quality in Use.

The designed and developed system was called College Admission and Aptitude Test Decision Support System based on the gathered data, the following findings were made. The security of the system was done through the use of provided username and password of the system administrator and student's user was accepted by the respondents in terms of all criteria presented. The system databases are well functioning and was successfully developed displaying student's profile that was done through the provided registration form and students find their individual records using search bar. Student's results on the examination were automatically generated by the system after taking of exams and can provide printout copy for the students. It can also provide administrators with statistical records in graphical display as to how many percent of students passed and failed in the exam.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

- Students examination
- Freshmen students
- Examination result

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Preparation for one's future is one of the many concerns of our youth today. As young adults, they are now more conscious about the real situation on the level of competitiveness in different professions in today's diverse industry. Pressure is piled on young people to choose qualifications for a career they want. Universities often demand top grades for certain courses. With students choosing careers that may not suit them, a worrying scenario could emerge: they could be stuck in careers like round pins in square holes. This is detrimental for the students because reconsidering a career is an enormous decision. But it is also detrimental for employers who will struggle to keep excellent students interested in careers that they no longer want. Choosing the right career that suits one's interests, abilities and skills is very essential. It is through this crucial decision that one's future will soon depend.

Times have changed and the decision-making process in the new age is entirely different from the past and significantly more challenging. Businesses today need to be agile and react in real-time in order to compete in the dynamic digital economy. The influx of data can either make or crush strategic decision-making. Repercussions of a bad decision can now be even more damaging in today's highly transparent world (Fernandez, 2016). Same thing is true in deciding one's career in considering one's future. And before all these happen, as a student, one has to be sure of what he is entering into: deciding on what course to take in college.

Entrance examination in the chosen university is one of the factors students consider in choosing their course. Through its result, they may be able to know which course is most suitable for them. As students themselves, the researchers decided to design a software that could help the system in giving entrance examination in Capiz State University, Pontevedra. Through this software, the students may decide on which course is suitable for them based on the examination result they received. And since this is a decision support system, this could also be helpful for the guidance counselor in assisting incoming freshmen students in making career or college plans (Fernandez, J. 2016).

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(11) Materials and Methods or Methodology (This includes the materials, methods,duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

MATERIALS

The materials to be used in conducting this study and in creating the College Admission and Aptitude Test Decision Support System are as follows,

- Laptop or Personal Computer

HARDWARE SPECIFICATIONS

Hardware

Computer or Laptop
Memory (RAM)
Hard Disk Space

Minimum Requirements

4 GHz minimum, multi-core processor
At least 4GB or Higher
Recommended 500 GB or Higher

SOFTWARE SPECIFICATIONS

Software

Brand
Product Line
Processor
Min Hard Drive Space
Min RAM Size
Min Operating System

Version

Google
Google Chrome Web Browser
Pentium 4 processor for higher
100MB
128MB
Windows 10

SOFTWARE DESIGN MODEL

Software Development Life Cycle (SDLC)

Iterative Development Model

In Iterative Model, iterative process starts with a simple implementation of a small set of the software requirements and iteratively enhances the involving version until the complete system is implemented and ready to be deployed. At each iteration, designed modifications are made and new functional capabilities are added. The basic idea behind these methods is to develop a system through repeated cycles and in smaller portion at a time. Following is the diagram for the software Development Life Cycle.

Below is the visualization of the Iterative Model to be used in this study.

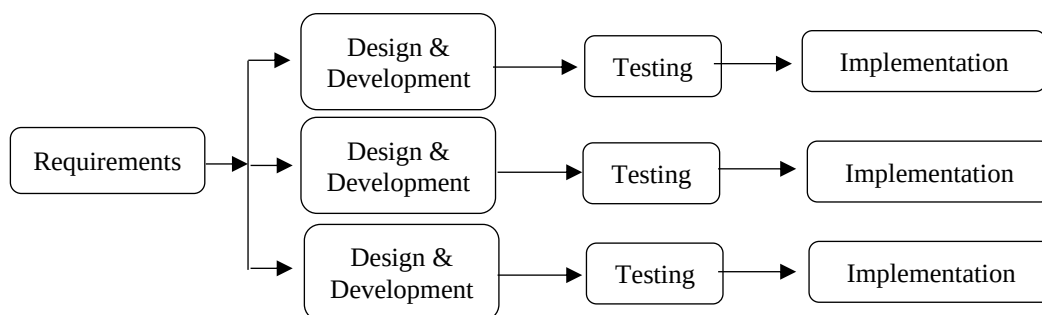


Figure 1. Iterative Model of the College Admission and Aptitude Test Decision Support System

Software Development Phases

Requirement Gathering & Analysis: In this phase, requirements are gathered from customers and check by an analyst whether requirements will fulfil or not. Analyst checks that need will achieve within budget or not. After all of this, the software team skips to the next phase.

Design: In the design phase, team design the software by the different diagrams like Data Flow diagram, activity diagram, class diagram, state transition diagram, etc.

Testing: After completing the coding phase, software testing starts using different test methods. There are many test methods, but the most common are white box, black box, and grey box test methods.

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Implementation: In the implementation, requirements are written in the coding language and transformed into computer programs which are called Software.

Deployment: After completing all the phases, software is deployed to its work environment.

Maintenance: The software is monitored to ensure it continues to function as it was designed to, and repairs or upgrades are performed as needed. The maintenance phase occurs once the software is fully operational.

CONCEPTUAL FRAMEWORK

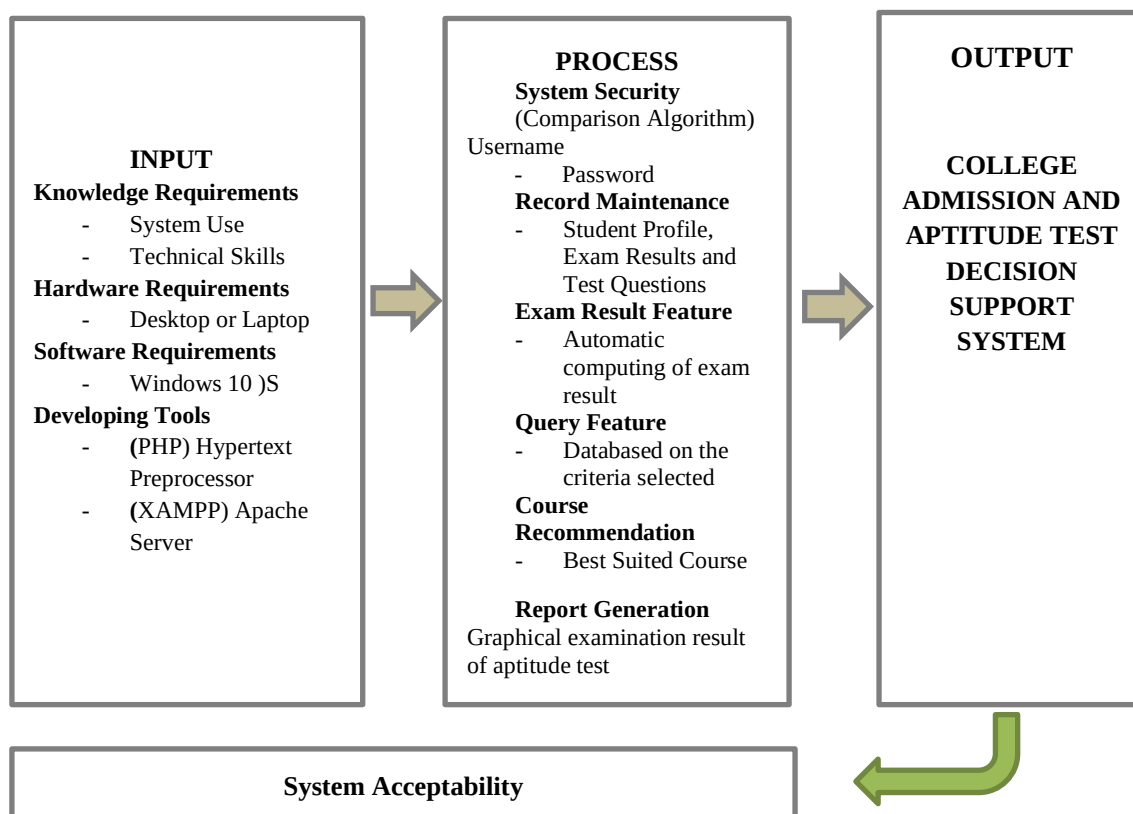
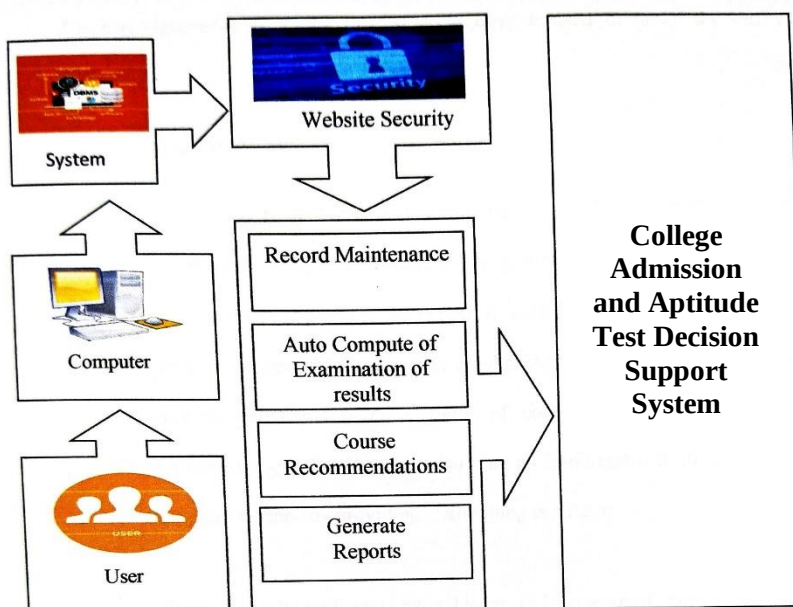


Figure 2. Conceptual Framework (IPO) of the College Admission and Aptitude Test Decision Support System

SYSTEM ARCHITECTURE



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Figure 3. System Architecture of the College Admission and Aptitude Test Decision Support System

Instruments

The system was tested for final evaluation or acceptability by the end users using the ISO 92010. It was evaluated in terms of product quality such as functional suitability, performance efficiency, compatability, usability, reliability, security, maintainability and portability, and quality in use such as effectiveness, efficiency, satisfaction, freedom from risk and context coverage.

The acceptability of the College Admission and Aptitude Test Decision Support System were rated by the respondents using the scoring below:

Range of Score	Verbal Interpretation	Description
4.5 – 5.0	Highly Accepted	Respondent agrees to all the features of the system.
3.5 – 4.49	Accepted	Respondent agrees to about 80% of the system's features.
2.5 – 3.49	Moderately Accepted	Respondent is satisfied with he features it offers.
1.5 – 2.49	Slightly Accepted	Respondent is slightly beginning to dislike some of its features
1.0 – 1.49	Unaccepted	Respondent doesn't accept all the features of the system

Statistical Tools

A total of 30 respondents viewed the website. After viewing, the questionnaire was distributed to them. To make data analysis easier, a tally sheet was provided where respondents were recorded. To avoid repetition of encoding, data were tabulated for analysis. The statistical measures used for the acceptability were the mean and standard deviation.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Product Quality

The acceptability level of College Admission and Aptitude Test Decision Support System in terms of all aspects in product quality which consisted of functional suitability, performance efficiency, compatability, usability, reliability, security, maintainability, and portability got a grand mean of 4.85 which can be interpreted as *Highly Accepted*. Meaning the respondents agreed in terms of product quality of the system.

Product Quality	Mean Average	Verbal Interpretation
1. Functional Suitability	4.90	Highly Accepted
2. Performance Efficiency	4.83	Highly Accepted
3. Compatability	4.78	Highly Accepted
4. Usability	4.88	Highly Accepted
5. Reliability	4.79	Highly Accepted
6. Security	4.87	Highly Accepted
7. Maintainability	4.87	Highly Accepted
8. Portability	4.87	Highly Accepted
GRAND MEAN	4.85	Highly Accepted

Table 1. Summary of Results in Product Quality

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Quality in Use

The quality in use which consist of effectiveness, efficiency, freedom from desk and context coverage got a grand mean of 4.9 which can be interpreted as *Highly Accepted*. Meaning the respondents agreed in terms of quality in use of the system.

Product Quality	Mean Average	Verbal Interpretation
1. Effectiveness	4.90	Highly Accepted
2. Efficiency	4.90	Highly Accepted
3. Satisfaction	4.88	Highly Accepted
4. Freedom from Risk	4.82	Highly Accepted
5. Context Coverage	4.82	Highly Accepted
GRAND MEAN	4.86	Highly Accepted

Table 1. Summary of Results in Quality in Use

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusion

Based on the gathered data, the following conclusions were drawn:

1. The system has an information security feature using username and password that can secure data to avoid being access by unauthorized personnel.
2. A record maintenance feature was provided using database to store all the information needed by the user.
3. The system has an examination result feature that will automatically compute examination result.
4. The system has a report query feature based on selected criteria.
5. The system has a recommendation feature that recommends the most relevant course for the students, based on examination results.
6. The system has a report generation feature using graphical presentation of result.
7. The acceptability of the study in terms of Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, Portability, Effectiveness, Efficiency, Satisfaction, Freedom from Risk and Context Coverage is "Highly Accepted." This implies that the respondents generally agreed on all aspects of its acceptability.

Recommendations

Based on the development and implementation of the College Admission and Aptitude Test Decision Support System, the following recommendations can be made:

1. The system may be used for better convenience and effectiveness of the university student entrance exam and generating report information which is essential to the office of the student's affairs and colleges.
2. Similar study may be conducted in the future students services methods and to improve/add additional features in the system to expand its scope and make it more beneficial to administrators, faculty & staff and students.
3. Further studies about the acceptability of College Admission and Aptitude Test Decision Support System may be conducted using other parameters aside from functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, portability, effectiveness, efficiency, satisfaction, freedom from risk and context coverage.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

1. **Information Security.** Sign-in form was provided to meet the security of the system. To use this, the user is required to enter username and password before they can enter his/her account page. The security interface was made through the use of Hypertext Preprocessor (PHP) Language.
2. **Record Maintenance.** Database was provided to store information such as, profile, exam results and test question provided by the system.
3. **Exam Result Feature.** The system has an automatic checking and computation of examination result.
4. **Query Feature.** The designed application has a feature that will filter the needed data based on the criteria selected.
5. **Course Recommendation.** The system has a feature that can recommend the best suited course for the

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student based on the high scores of obtained from the examination taken.

6. **Report Generation.** The system used a graphical representation of the result of the examination taken by the students.

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

A technology transfer activity design of the College Admission and Aptitude Test Decision Support System as a research output of the campus could be designed as follows:

1. Pre-deployment survey: Conduct a pre-deployment of the different colleges of the university to identify the specific technology adoption related to student services. This can be done through installation of the system to a certain college/department and office of the student affairs
2. Development of the Decision Support System: Based on the identified needs, design and develop the College Admission and Aptitude Test Decision Support System. This could involve hiring IT professionals and web developers to enhance the system and the necessary features.
3. Training and capacity building: Once the College Admission and Aptitude Test Decision Support System has been developed, conduct training and capacity-building activities for university staff, including the IT team, department heads, and front-line staff. This will ensure that they have the necessary skills to operate and maintain the system.
4. Pilot testing: Conduct a pilot test of the College Admission and Aptitude Test Decision Support System to identify any glitches or issues that need to be addressed. This can be done by selecting a small group of students to use the system and provide feedback.
5. Launch and dissemination: Once the College Admission and Aptitude Test Decision Support System has been fully developed and tested, launch it to the entire university community. This can be done through an official launch event, promotional materials, and targeted communication to different departments and units.
6. Maintenance and updates: Ensure that the College Admission and Aptitude Test Decision Support System is regularly maintained and updated to keep it running smoothly and efficiently. This will involve regular checks for bugs, updates to security features, and improvements to functionality based on user feedback.
7. Evaluation and feedback: Finally, conduct an evaluation of the College Admission and Aptitude Test Decision Support System to assess its impact and effectiveness. This can be done through surveys, focus group discussions, and usage data analysis. Use this feedback to improve the system further and to inform future technology transfer activities.

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

SOURCES OF EXPENSES	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	₱10,000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)	₱20,000.00
Total	₱30,000.00

Table 3. Budgetary Requirements for designing the College Admission and Aptitude Test Decision Support System

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

This study was undertaken with the aim of achieving a substantial benefit. It was made possible through the combined efforts of many individuals who worked tirelessly, persevered through challenges, collaborated effectively and remained determined to see the project through to its successful completion. Their collective contributions were instrumental in bringing this work to fruition and realizing its potential for significant gain.

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The researcher is truly blessed to have a wonderful support system throughout this journey, and they dedicate this thesis to all those who have helped them along the way.

To their beloved parents who served their refuge in their ups and downs, also for the loved and understanding, moral, financial and spiritual, and for their worthwhile pieces of advice, and hardships in the pursuit of this endeavor;

To all the faculty and staff of Computer Department for the support, consideration and the encouragement especially the knowledge shared to the researchers;

They would also like to thank their colleagues and friends for their encouragement and support. Their contributions have been instrumental in the success of this project.

Finally, our earnest thanks to the Almighty God who give them all this people as blessing, who blessed them with wisdom, strength, perseverance, the courage to survive despite all the challenges, sacrifices and adversities. Thank you very much!

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

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